COMP3751 VR Coursework

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Q3.3

Additional Accelerometer Processing

Before correcting tilt I applied two preprocessing steps to the raw accelerometer signal. Firstly I added magnitude gating to reject samples with vector magnitude outside 0.85–1.15g. As LaValle et al. noted, the accelerometer measures the vector sum of gravity and linear acceleration, so readings deviating significantly from $9.8\,\mathrm{ms}^{-2}$ are more likely caused by dynamic motion than tilt [1, 3]. Madgwick also recommends reducing accelerometer influence during periods of large detected acceleration [2]. Second, I applied an exponential moving average (EMA) per [4] with factor 0.1 as a low-pass filter to reduce high-frequency noise in the accelerometer signal before it reaches the complementary filter.

Investigating Alphas in Complementary Filter

Figure 1 shows my complementary filter ran over five separate alphas (accelerometer trust over gyroscope) plotted as Euler angle traces. Yaw is excluded for reasons discussed in Q6.2. The dead reckoning baseline uses the same method without accelerometer correction (alpha=0).

Firstly, the chaotic spikes roll spikes (top panel) between ∼11–16s are just a mathematical artefact of Euler angle representation due to the conversion reaching a singularity and roll and yaw values swapping unpredictably [2]. During the first rotation phase (∼0–7s), all filters track virtually identically as rotation is slow enough that just gyroscope integration is sufficiently accurate and drift hasn't yet accumulated.

The most notable region is the final rest period (∼23–27s), where all angles should return to 0°. In the roll graph, dead reckoning settles highest (approximately +25°) due to uncorrected gyroscope drift. Low alpha values (0.0005, 0.001) provide modest correction, sitting slightly below dead reckoning, and higher values (0.01, 0.02, 0.1) correct more aggressively, actually overcorrecting a bit dipping below 0°. All lines follow a shared dip at ∼23s before rising into the rest period. However, looking at the trace quality, there's a clear trade-off: higher alpha values return closer to zero but with noticeably spikier lines because the filter trusts the accelerometer significantly at a time where short-term readings are unreliable post-rotation [1].

The pitch graph shows a similar but messier result. From ∼21–23s all filters rise above zero before dropping below it at ∼23s as the Z-rotation finishes. Dead reckoning then settles furthest from zero at approximately −20°, with larger alpha values correcting that offset progressively more, though again at the cost of spikier traces reflecting the same short-term accelerometer noise [2]. Also, a notable crossover occurs at ∼25s where the low alpha traces which were fractionally below dead reckoning rise above it, possibly due to accumulated correction debt from sustained under-correction and then greater correction applied once a stable gravity signal is available at rest.

Overall, the noise–drift trade-off is consistent across pitch and roll. Higher alpha reduces drift but amplifies filter sensitivity to small accelerometer errors. Alpha=0.02 gives the most balanced results, consistent with Madgwick's finding that there's a clear optimal gain that balances both drift and noise reduction [2]. It returns close to 0° in both roll and pitch without significant noise amplification, and avoids the correction debt crossover visible at lower values.

Q4.2

Additional Magnetometer Processing

Before applying yaw correction I implemented hard-and-soft-iron calibration, identified by LaValle as the two most significant sources of magnetometer error [3]. Hard-iron correction subtracts the midpoint of each axis's observed range from each reading, removing the constant offset introduced by ferromagnetic materials on the sensor's circuit board [5]. Without it, this offset shifts in a different direction relative to the sensor as the headset rotates, injecting a false yaw signal even though both the reference and live readings are affected.

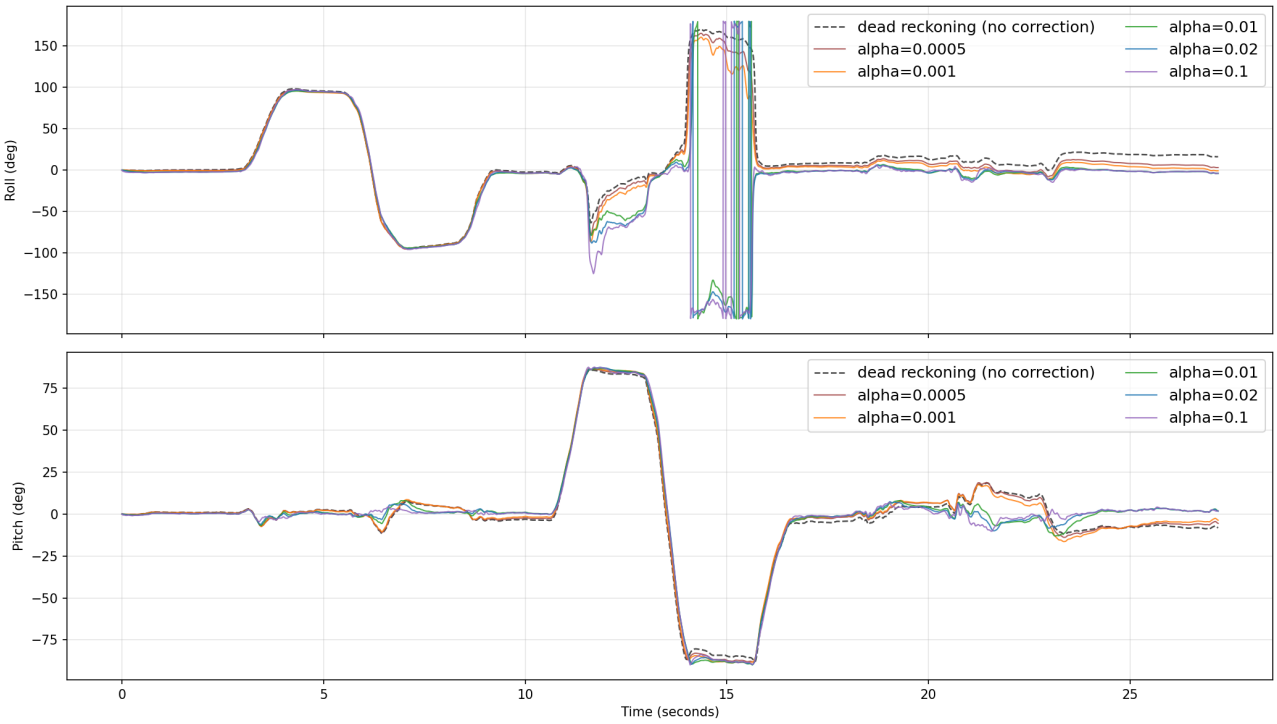


Figure 1: Traces showing the affect of the complementary filter alpha on roll and pitch over time.

Soft-iron correction then rescales each axis by its own range, compensating for the fact that nearby materials cause the sensor to report different field strengths depending on which way it is pointing [5, 6]. Without it, the same physical rotation produces inconsistent readings. These calibrations were applied identically to the $t = 0$ reference and each subsequent reading, ensuring yaw error is always computed in a consistent calibrated space.

A magnetometer EMA (factor 0.1) was applied to the corrected readings for the same noise-reduction reasons as the accelerometer EMA. I also applied a dual reliability gate before firing any correction: a gyro rate gate suppressing correction when total angular rate exceeds 0.3 rad/s preventing the filter from chasing a stale smoothed heading during fast rotation, and a tilt gate suppressing correction when the sensor exceeds 20° from upright, reflecting LaValle's observation that the horizontal component of the magnetic field becomes unreliable at high inclination and may be too small to be useful [3].

Investigating Alphas in Magnetometer-Corrected Complementary Filter

Figure 2 shows yaw estimates produced by running the full fusion filter at five magnetometer-gain alpha values contrasted with the accel-only complementary filter as a baseline to show the true affect of magnetometer correction. Surprisingly it shows that the magnetometer correction doesn't actually improve on the accel-only baseline for the provided data, in fact making it actively worse by degrading the estimate at higher alpha values.

Interestingly, the accel-only filter already settles very close to 0 in the final rest period, meaning the gyroscope didn't actually accumulate much net yaw drift over the full sequence, so the magnetometer is correcting a problem that's nearly absent whilst introducing its own errors. This is consistent with Madgwick's findings that heading accuracy is inherently lower than pitch and roll accuracy because of the relatively small horizontal component of the Earth's magnetic field at typical test latitudes, and also aligns with the magnetometer having a lower bandwith than the gyroscope and accelerometer, which makes it much less reliable under the dynamic conditions this sequence contains [2].

I conclude that this appears to be a reference quality issue from the initial rest period (∼0–3.5s), whereby both reliability gates are passed for a significant number of frames, activating many corrections before rotation even begins. The reference vector is the singular $t = 0$ magnetometer sample but all subsequent live readings are EMA-smoothed and converge towards the mean of the first few seconds of data, which is more accurate than the single noisy sample. The filter is therefore repeatedly correcting away from the more accurate smoothed signal towards the noisier reference point, accumulating a small but persistent positive yaw offset that scales with alpha and propagates throughout the sequence. It's negligible at small alpha (0.0005 and 0.001) but reaches several degrees at alpha=0.1, and once introduced it compounds through subsequent phases without a self-correcting mechanism.

This initial offset then makes the Z-rotation phase worse, as even with the strict gate threshold, a small number of frames fire correction at temporarily slow moments in the rotation, at which point the magnetometer EMA reflects an average over the sensor's recent rotational trajectory instead of its current orientation, which is a lagged blend of where the sensor's been, not where it is. Rotating this stale average to the global frame gives a heading that is neither the current world-frame direction nor a useful reference, so the resulting corrections push the estimate away from the true rotation value. This produces the visible overshoot: the accel-only baseline peaks at approximately 105° during the positive Z-rotation while alpha=0.1 peaks at approximately 120°, with this deviation scaling with higher alpha. The negative Z-rotation shows the mirror effect, with higher alpha values reducing the magnitude of the peak, bringing alpha=0.1 to approximately −70 compared to the accel-only −90. In the final rest period, where the accel-only filter and the two lowest alpha values settle close to 0°, alpha=0.01 and 0.1 are elevated at approximately 20 and 25 respectively, reflecting the accumulated result of corrections having fired in the wrong direction during the rotation phases.

Madgwick mentions there's a clear optimal gain value for the MARG filter, balancing drift correction against the noise introduced at each correction step [2]. Here however, I don't believe there is such a value, as the error introduced by each correction exceeds the drift being corrected. The lowest tested values (0.0005 and 0.001) cause the least damage and remain close to the accel-only baseline, consistent with Madgwick's recommendation that the magnetometer gain be kept substantially lower than the accelerometer gain given its higher noise characteristics.

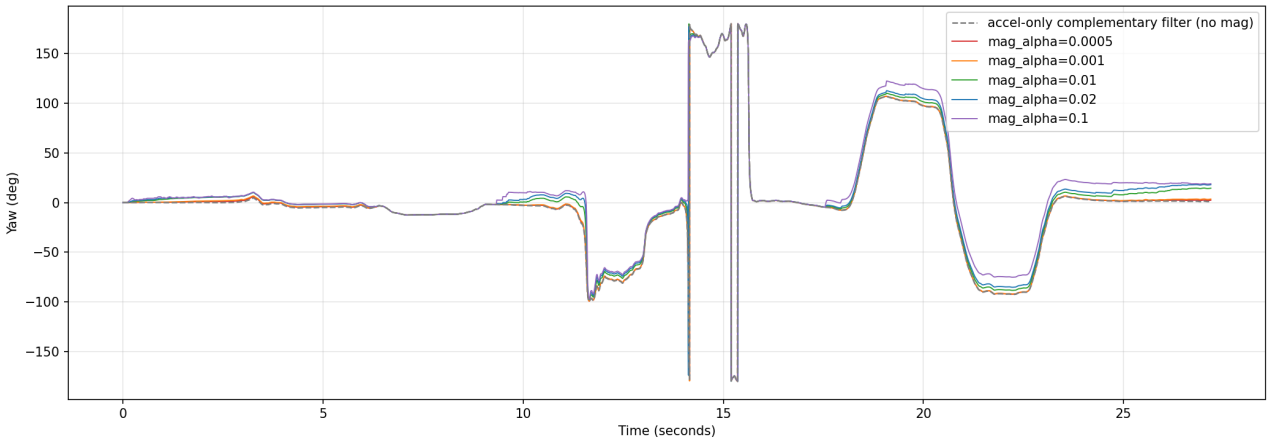


Figure 2: A trace showing full fusion filter affect of alpha on yaw over time.

Q6.1: Static Renders

Figure 3 shows static renders of the bunny produced by the complementary filter (alpha=0.02): the rest frame (t=0), and the frames at approximate peak roll, pitch and yaw in the IMU sequence. They confirm that my transformation matrices and fused orientation estimate function correctly, as each render reflects the expected axis of rotation.

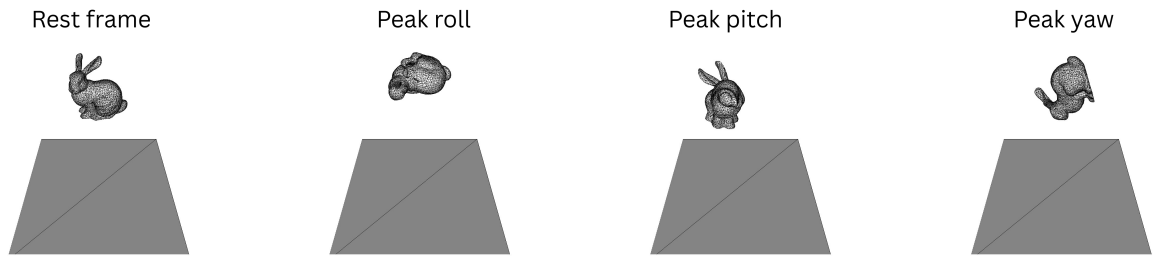


Figure 3: Static renders showing the bunny at rest and peak roll, pitch and yaw.

Q6.2: Comparing Dead Reckoning with and without Accelerometer

Figure 1 from Q3.3 presents the direct comparison of simple dead reckoning vs the accelerometer-fused upgrade at varying alpha. Table 1 summarises key outcomes during the final rest period (∼23–27s), the most revealing phase as both pitch and roll should return to 0. Alpha=0.02 was used and root mean square (RMS) was chosen as the metric as it penalises large drift more than constant small offsets. Thus looking at the table, it becomes apparent that utilising the accelerometer data for corrections significantly reduces this drift, demonstrated by a much lower RMS error in both roll and pitch.

Table 1: RMS error from 0 during final rest period (23–27.5s)

Filter	**Roll RMS (deg)**	**Pitch RMS (deg)**
Dead reckoning (gyroscope only)	18.6	8.3
Complementary filter (alpha=0.02)	2.8	3.1

Advantage of dead reckoning: clean tracking during active rotation

As dead reckoning just integrates gyroscope data, it produces a consistently smooth trace throughout the sequence. By contrast, the complementary filter continuously blends accelerometer signal, which is an inherently unreliable short-term reference. This is because it measures the vector sum of gravity and all linear accelerations acting on the sensor, so any movement, whether that be deceleration, vibration or sudden heading changes, temporarily corrupts the gravity estimate [1]. This is visible in Figure 1, where higher alpha traces are spikier due to persistent noise whereas dead reckoning and low alphas are smoothed throughout all rotation phases. Higher alphas cause each corrupted sample to be applied as tilt correction before the EMA can dampen it, which dead reckoning is immune to as it never touches the accelerometer.

Disadvantage of dead reckoning: drift

Without a corrective reference, gyroscope integration errors consistently accumulate. LaValle terms this drift error, which requires additional sensors for detection and compensation [3]. The findings clearly quantify this: dead reckoning finishes with an 18.6° roll RMS error and 8.3° pitch RMS error during the rest period, drifting approximately +16 in roll while the sensor is physically stationary at its starting orientation. The complementary filter reduces these to 2.8° and 3.1° respectively with incremental accelerometer-derived tilt corrections. Madgwick describes this trade-off as a filter gain problem, whereby the gain must be high enough to correct integral drift but low enough that correction-sensor noise isn't over-amplified [2], as explored in Q3.3.

Shared limitation: yaw

However, neither approach can correct yaw drift because the cross product of estimated and measured gravity vectors gives zero rotation signal about the vertical axis so no tilt correction signal is available [1].

Q6.3: Performance of both methods

6959 samples are processed per filter, and the cost of each operation per sample contributes to overall computational cost, with multiply/divide and addition/subtraction being equal cost. Table 2 summarises the calculated arithmetic operations used for each method, which I calculated by tracing the function calls each filter makes (full workings in my submission).

Table 2: Estimated arithmetic operations per sample

Filter	**Mult/Div**	**Add/Sub**	**Trig**	**Sqrt**
Dead reckoning	38	17	6	0
Complementary filter (additional)	102	62	5	5
Complementary filter (total)	140	79	11	5

Per sample, dead reckoning calls three functions: three multiplications for $\omega \cdot \Delta t$ on each axis, then `convert_euler_to_quaternion` (3 half-angle multiplications, 6 trigonometric evaluations, and 16 multiplications plus 4 additions to build the four quaternion components), then one Hamilton product to accumulate the rotation (16 multiplications, 12 additions). Dead reckoning doesn't include a normalisation step, so quaternion

magnitude drifts numerically over time, and significantly, without any corrective reference, gyroscope bias orientation errors accumulate unbounded. In total, I calculate approximately 38 multiplications/sample, which would be roughly 264,000 across the full dataset.

The complementary filter does all that then adds eight more steps per sample. First, it normalises the gyro estimate (8 mult, 1 sqrt), then applies a low-pass EMA filter across the three accelerometer axes (6 mult, 6 add), a magnitude gate square root, two full quaternion sandwich products to rotate the accelerometer reading into the global frame (32 mult, 27 add), a cross product and two further normalisations to compute tilt axis (21 mult, 2 sqrt), building the correction quaternion (8 mult, 4 trig), applying it via another Hamilton product (16 mult, 12 add) and a final normalisation (8 mult, 1 sqrt). Overall that's approximately 102 more multiplications/sample, reaching 144 multiplications/sample and roughly 974,000 across the full dataset. Therefore it's approximately 3.7× more multiply-intensive than dead reckoning.

To confirm this, I timed both filters over 10 runs for stability, and my results are shown in Table 3. Dead reckoning completed in 318.1 ms (45.7 μs/sample) and the complementary filter in 504.2 ms (72.4 μs/sample), giving a measured wall-clock ratio of 1.59×. Although lower than my calculated 3.79× multiply ratio, I consider this expected given memory accesses and interpreter costs; importantly, it still reveals the complementary filter has a non-negligible cost increase.

Table 3: Measured filter execution time (mean over 10 runs)

Filter	**Total (ms)**	**Per sample (μs)**
Dead reckoning	318.1	45.7
Complementary filter	504.2	72.4
Runtime ratio (complementary filter / dead reckoning)		1.59×
Calculated mult-op ratio (complementary filter / dead reckoning)		3.68×

Therefore I conclude that the additional cost of accelerometer fusion is negligible in absolute terms. IMUs are useful exactly for their high sampling rates [3], and both methods here are well within latency requirements for comfortable VR. LaValle notes motion-to-photon latency in modern systems is approximately 15–25 ms [3] and both filters consume under 0.1 ms, leaving the vast majority of the latency budget available for rendering and display. Therefore whilst this overhead negligible on modern hardware, the 1.59× processing time increase remains relevant for resource-constrained systems.

Q6.4: Positional Tracking

On paper, positional tracking could be added. This would be done by double-integrating the accelerometer readings after subtracting the gravity component derived from the current orientation estimate, which gives velocity on the first integration and position on the second [3]. However, LaValle identified that as impractical, because whereas gyroscope calibration error produces linearly growing drift through a single integration, double integration causes calibration error to quadratically accumulate, which is intolerable after a very small amount of time [3]. Also, the true body acceleration can't be accurately separated from gravity, especially during fast rotation [3]. The IMU alone doesn't have a mechanism to detect or correct accumulated positional drift, and in fact can't even distinguish motion at constant velocity from rest [3]. Thus I conclude that positional tracking from this data is impractical without an external absolute position reference like cameras.

Q6.5: Collision Detection with Spheres

Whilst computationally inexpensive, sphere-based collision is fundamentally flawed as it can't encapsulate non-spherical geometries well and the collision radius requires subjective manual tuning.

Analysing the normalised bunny mesh showed that quantitively. The model spans ±0.69 in X, ±0.68 in Y, and has a true minimum bounding sphere radius of 1.037 from the mesh centroid. Both radii used in the simulation (0.20 for the stationary bunny, 0.12 for the falling ones) are smaller than that, so collisions are only detected when the bunny centres fall within 0.32 units of each other, where meshes could already have visually connected. If I set the radius higher, or even to the true bounding radius of 1.037, it would make collisions appear more unnatural as bunnies wouldn't necessarily visually connect when a collision is triggered.

I also calculated bunny's aspect ratio, revealing a relatively cubic shape, with height/width ≈ 0.99 and height/depth ≈ 1.28. Therefore a sphere a reasonable shape approximation but much of its volume is just empty space whereby collisions could still trigger without visual overlap.

In this scene, each of the n falling bunnies is checked only against the fixed central bunny, giving $O(n)$ per frame, but in a general scene with all-pairs checking this would become $O(n^2)$. A method like axis-aligned bounding boxes (AABBs) would reduce false positives at the same O(1) per test cost through tighter geometric

fitting to reduce false positive collisions [7]. For general all-pairs scenarios, a bounding volume hierarchy (BVH) would reduce that $O(n^2)$ to O(nlogn) construction with O(logn) per-query traversal, though the upfront build cost and memory overhead could become significant at scale [7].

References

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